

What is the Relational Model?

Definition

The Relational Model is a logical model for organizing data into relations (tables) and defining relationships between them using mathematical principles from set theory .

Example:

Students

StudentID	Name	DepartmentID
101	Aadya	10
102	Arpit	20

Departments

DepartmentID	DepartmentName
10	CSE
20	ECE

The relationship is:

```
Student
  |
DepartmentID
  |
Department
```

3. The Problems with Older Database Models

A. Hierarchical Database Model

Company

|

|— HR

| |— Employee A

| |— Employee B

|

|— Engineering

 |— Employee C

 |— Employee D

each child had one parent node , now what if employee c works for for hr and engineering , this leads to data redundancy , since two parents were not allowed in heirarichal databass .

B. Network Database Model

To solve that,
network databases allowed:
Multiple parents.

aadya

/

software engineer vlsi engineer .

Although more flexible,
they became complicated.
Applications had to know:

- Where data was stored
- Which path to follow

Developers spent significant effort navigating data .

The relational model is primarily based on:

- Set Theory
- First-Order Predicate Logic .

A. Set Theory

A set is:

- An unordered collection of distinct elements.

Example:

```
{1, 2, 3, 4}
```

Notice:

- No duplicates
- No ordering requirement .

Although SQL implementations allow duplicate rows unless you explicitly remove them, the original relational model treats a relation as a mathematical set of tuples.

B. Predicate Logic

Predicate logic allows statements like:

```
Student CGPA > 9
```

SQL:

```
SELECT *  
FROM Students  
WHERE CGPA > 9;
```

The WHERE clause is essentially expressing a logical predicate that each row must satisfy. Predicate logic allows you to model complex real-world facts that cannot be expressed using simple true/false sentences. Its foundational elements include: [1]

Predicates: The properties or relations of objects. For example, in the statement "x is a prime number," "x" is the object and "...is a prime number" is the predicate, often denoted as $P(x)$. [1, 2] .

Relational Database = Database with relationships.

Wrong.

The word relation is a mathematical term from set theory.

A relation is a set of tuples sharing the same attributes.

10. Advantages of the Relational Model

1. Simple Structure

Everything is represented using tables.

Easy to understand.

2. Data Independence

Applications focus on what data they need.

The DBMS decides how to retrieve it.

3. Reduced Redundancy

Supports normalization,
which minimizes unnecessary duplication.

4. High Data Integrity

Constraints help ensure valid data.

Examples:

- Primary Key
- Foreign Key
- CHECK
- UNIQUE

5. Powerful Query Language

SQL enables complex queries without requiring developers to manually navigate data structures.

Limitations of the Relational Model

No technology is perfect.

1. Fixed Schema

Changing table structures requires planning and migrations.

2. Complex Object Storage

Nested JSON, graphs, and deeply hierarchical structures are often less natural than in specialized databases.

3. Horizontal Scaling

Scaling relational databases across many machines is generally more challenging than scaling some NoSQL systems.

4. Joins Can Become Expensive

Large joins across massive datasets may require significant computation, especially without proper indexing.

They remain the backbone of banking, finance, healthcare, ERP systems, and much of modern backend infrastructure.